

several maker boards such as the Arduino and the Raspberry Pi. There are several projects which are new and those which have been showcased in earlier editions which you can download or view from <http://dgit.in/MakerTechFTResources>. Here are a few of the most interesting maker projects you can get your hands on.

Remote controlled lights

Turning on the lights and turning them back off are things we do throughout the day. Many a time, we forget to switch off electronic appliances when we leave home. It's all good until you forget to switch off your air conditioner or television set, which can effortlessly waste electricity and add to light bill costs. For the purpose of this DIY, we'll look at a simple circuit and code that will help you switch a lamp on and off. This is an extension to the LED blinking program that we saw in the previous chapter.

Hardware needed:

- ♦ Raspberry Pi
- ♦ Relay circuit
- ♦ Lamp for circuit testing

If you're a beginner, we recommend using a relay circuit from the market. You could also build your own relay circuit using a motor driver or a transistor. In case you want to build your own relay circuit, please take the help of a friend who has adequate knowledge about electronics. Alternatively, you can follow an online tutorial such as the one available at <http://dgit.in/1098eyK>.

Plant watering System

In this DIY, we'll make an automated plant watering system. This will allow you to water the plants in your home, building or even colony while sitting in another part of the world. For this purpose, we'll create a web server that will listen to a request for switching the motor ON and OFF.

Hardware needed:

- ♦ Raspberry Pi Breadboard
- ♦ Water pump / Solenoid valve
- ♦ Solid State Relay
- ♦ N3904 Transistor
- ♦ 4.7 k ohm Resistor